Chapter 3. Set theory and discrete probability

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3.1 Elements, subsets, and extensionality

Sets retain neither repetitions nor order, so $\{1, 2, 2\} = \{2, 1\}$. Extensionality states

$$A = B \iff \forall x(x \in A \leftrightarrow x \in B).$$

The inclusion $A \subseteq B$ means $\forall x(x \in A \Rightarrow x \in B)$; proper inclusion $A \subsetneq B$ also requires $A \neq B$. Vacuous truth gives $\varnothing \subseteq A$. Membership $\varnothing \in A$ requires the empty set actually to be an element.

For $A = \{\varnothing, \{1\}\}$, there are two elements, with $\varnothing \in A$, $\{1\} \in A$, but $1 \notin A$. Also $\{\varnothing\} \subseteq A$, whereas $\{1\} \not\subseteq A$ since $1 \notin A$. Braces change the object being discussed.

Set equality is equivalent to the two inclusions. One $x \in A \setminus B$ refutes $A \subseteq B$.

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3.2 Powersets contain subsets

Define $\mathcal{P}(A) = \{S : S \subseteq A\}$. For $A = \{a, b\}$,

$$\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}.$$

When $|A| = n$ is finite, subsets biject with n-bit membership vectors, giving $|\mathcal{P}(A)| = 2^n$. In particular $\mathcal{P}(\varnothing) = \{\varnothing\}$ has one element.

For any A, B,

$$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B),$$

since membership on the left means $S \subseteq A \cap B$, equivalently $S \subseteq A$ and $S \subseteq B$, exactly membership on the right.

Generally $\mathcal{P}(A \cup B) \neq \mathcal{P}(A) \cup \mathcal{P}(B)$. With $A = \{a\}, B = \{b\}, a \neq b$, the mixed subset $\{a, b\}$ belongs only to the left. The right side is always included in the left.

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3.2.1 Gray code changes one element at a time

Membership vectors can follow reflected Gray-code order. Starting with $G_0 = (\epsilon)$, define

$$G_n = (0G_{n-1},\ 1\,\mathrm{reverse}(G_{n-1})).$$

Prefixes distinguish the two halves. Inductively, each half contains all suffixes without duplication, so together they list all 2^n vectors.

Positions	Membership vectors for three elements	Subsets for bits a,b,c
0,1	000,001	$\varnothing, \{c\}$
2,3	011,010	$\{b, c\}, \{b\}$
4,5	110,111	$\{a, b\}, \{a, b, c\}$
6,7	101,100	$\{a, c\}, \{a\}$

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3.2.1 Gray code changes one element at a time (continued)

Within each half, adjacent suffixes differ in one bit. At the join, suffixes both equal the last member of G_{n-1}, and only the prefix differs. The first vector is 0^n and the last 10^{n-1}, so they also differ in one bit for $n \geq 1$. This cyclic adjacency order differs from increasing binary numerical order. Karnaugh maps later use this property.

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3.3 Set operations and a fixed universe

Fix a universe U, with all sets subsets of it:

$$
\begin{aligned}
A \cup B &= \{x \in U : x \in A \lor x \in B\},\\
A \cap B &= \{x \in U : x \in A \land x \in B\},\\
A \setminus B &= \{x \in U : x \in A \land x \notin B\},\\
A^c &= U \setminus A,\\
A \triangle B &= (A \setminus B) \cup (B \setminus A).
\end{aligned}
$$

Symmetric difference retains elements belonging to exactly one set and corresponds to XOR.

Complements depend on the universe. For $A = \{1, 2\}$, its complement in $\{1, 2, 3\}$ is $\{3\}$, but in $\{1, 2, 3, 4\}$ it is $\{3, 4\}$. An unspecified universe leaves a complement expression incomplete.

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3.4 Set laws and their duals

Property	Union form	Intersection form
Commutativity	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Idempotence	$A \cup A = A$	$A \cap A = A$
Identity	$A \cup \varnothing = A$	$A \cap U = A$
Bounds	$A \cup U = U$	$A \cap \varnothing = \varnothing$
Absorption	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$
Distribution	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Complement	$A \cup A^c = U$	$A \cap A^c = \varnothing$
De Morgan	$(A \cup B)^c = A^c \cap B^c$	$(A \cap B)^c = A^c \cup B^c$

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3.4 Set laws and their duals (continued)

Translate arbitrary-element membership into Chapter 2's propositional equivalences to prove every law, rather than testing particular sets.

The paired laws are dual. Interchanging $\cup, \cap$ and $U, \varnothing$, retaining complements, exchanges the axioms with equally valid ones. Thus every universal identity derived from them has a valid dual. Accidental equalities of special sets do not qualify. For $A = \{1, 2\}, B = C = \{1\}, D = \varnothing$, $A \cap B = C \cup D$, but $A \cup B \neq C \cap D$.

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3.5 Element proofs of set equality

For arbitrary $x \in U$,

$$\begin{aligned} x \in A \cap (B \cup C) &\iff (x \in A) \wedge [(x \in B) \vee (x \in C)] \\ &\iff [(x \in A) \wedge (x \in B)] \vee [(x \in A) \wedge (x \in C)] \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

The middle step uses propositional distribution; extensionality then proves equality. Similarly,

$$x \in (A \cup B)^c \iff \neg[(x \in A) \vee (x \in B)] \iff x \in A^c \cap B^c.$$

A two-inclusion proof must take an arbitrary element in each direction. One specific Venn diagram alone does not establish the result for arbitrary sets.

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3.6 Indexed families and empty families

For $A_i \subseteq U, i \in I$,

$$\bigcup_{i \in I} A_i = \{x \in U : \exists i \in I, x \in A_i\}, \quad \bigcap_{i \in I} A_i = \{x \in U : \forall i \in I, x \in A_i\}.$$

If $I = \varnothing$, no existential witness exists, so the empty union is $\varnothing$. The universal condition is vacuous, so the empty intersection relative to fixed U is U.

Quantifier negation gives indexed De Morgan laws:

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \qquad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

For $A_n = \{1, \ldots, n\} \subseteq \mathbb{N}^+$, their union is $\mathbb{N}^+$, whereas their intersection is $\{1\}$. The existential and universal membership requirements differ.

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3.7 Cartesian products and partitions

Ordered pairs satisfy $(a, b) = (c, d) \iff a = c, b = d$. Define $A \times B = \{(a, b) : a \in A, b \in B\}$, of finite size $|A||B|$. Generally $A \times B$ and $B \times A$ are different sets; swapping coordinates supplies a bijection, not equality.

A partition $\{B_i\}$ has nonempty pairwise disjoint blocks with union U. Every element belongs to exactly one. The empty universe has the empty family as its partition; adding empty blocks is forbidden.

For $\{1, 2, 3, 4, 5, 6\}$, parity gives two blocks, and residues modulo three give three. Total probability also uses partitions, with additional positive-probability requirements for conditional denominators.

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3.8 Membership regions can be empty

Membership in three sets A, B, C is a three-bit pattern. For each $\epsilon \in \{0,1\}^3$, intersect each set or its complement according to that pattern, producing eight possible regions.

Each element has exactly one pattern. Regions are disjoint and cover U, but some may be empty. Only nonempty regions are partition blocks and atoms of the finite Boolean algebra generated by the sets.

For $A = B = C = \varnothing, U \neq \varnothing$, only region 000 is nonempty, rather than eight nonempty atoms. A Venn diagram may label geometric regions zero. Their shapes do not imply existence of elements.

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3.9 Indicators and inclusion-exclusion

Define $\mathbf{1}_A(x) = 1$ on A, zero elsewhere. Pointwise,

$$\mathbf{1}_{A\cap B} = \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A\cap B}.$$

Summing over finite U gives

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets, an element belonging to $k = 1, 2, 3$ sets receives coefficient $k - \binom{k}{2} + \binom{k}{3} = 1$; for $k = 0$, it receives zero. Hence

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

This pointwise proof does not depend on circle geometry. Chapter 8 extends it to arbitrary finite families.

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3.10 Checking all eight region counts

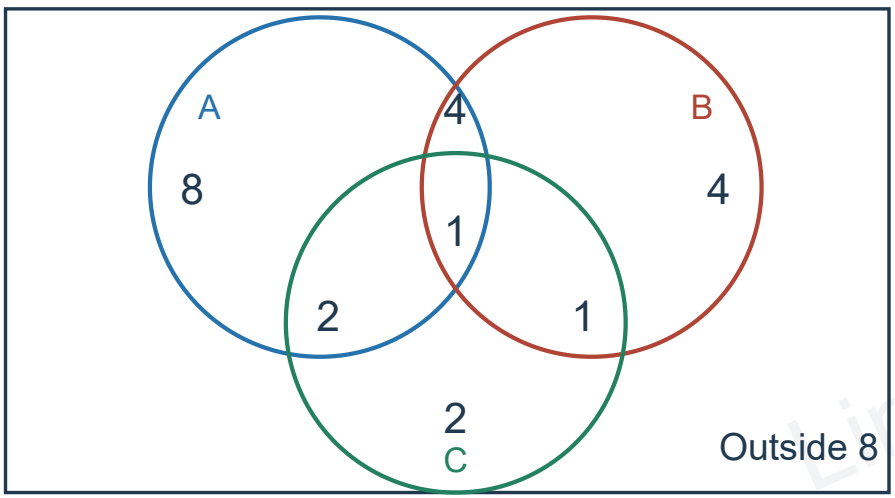
In $U = \{1, \dots, 30\}$, let A, B, C be multiples of two, three, and five. Their sizes are 15,10,6; pairwise intersections have sizes 5,3,2; the triple intersection contains only 30 and has size one.

Subtract the triple first, obtaining AB-only 4, AC-only 2, BC-only 1. Then A-only is 8, B-only 4, C-only 2, and outside 8.

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3.10 Checking all eight region counts (continued)

U={1,…,30}: A multiples of 2, B multiples of 3, C multiples of 5



|A|=15 · |B|=10 · |C|=6

|A∩B|=5 · |A∩C|=3 · |B∩C|=2

Triple =1; fill inside out, subtract overlaps.

Union: 8+4+2+4+2+1+1 = 22

Exactly one: 8+4+2 = 14

Region counts; areas are not proportional.

The union has 22 elements, exactly-one 14, exactly-two 7, and exactly-three 1. The check $14 + 7 + 1 + 8 = 30$ matches the universe and all regions are nonnegative.

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3.11 The discrete probability model

Fix a nonempty finite or countable sample space Ω, with outcome masses $p_\omega \geq 0$ summing to one. All subsets are events, with

$$P(A) = \sum_{\omega \in A} p_\omega.$$

Nonnegative sums can be rearranged, so disjoint unions add their masses. This proves normalization $P(\Omega) = 1$ and countable additivity.

Conversely, nonnegativity, normalization, and countable additivity yield this representation on countable Ω by setting $p_\omega = P(\{\omega\})$. No construction for arbitrary uncountable event spaces is needed here.

Only when finitely many outcomes are equally likely does $P(A) = |A|/|\Omega|$. A countably infinite space cannot give each singleton equal probability. Positive common mass makes the sum diverge; zero common mass sums to zero.

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3.12 Basic consequences of the axioms

Since $P(\Omega) = P(\Omega \sqcup \varnothing) = P(\Omega) + P(\varnothing)$, $P(\varnothing) = 0$. The partition $\Omega = A \sqcup A^c$ gives $P(A^c) = 1 - P(A)$.

If $A \subseteq B$, then $B = A \sqcup (B \setminus A)$ gives

$$P(B) - P(A) = P(B \setminus A) \geq 0,$$

proving monotonicity. Split two events into their exclusive parts and intersection to obtain

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

As $P(A \cup B) \leq 1$,

$$P(A \cap B) \geq P(A) + P(B) - 1,$$

also combining with nonnegativity. If $P(A) = 0.7, P(B) = 0.5$, their intersection is at least 0.2 without any independence assumption.

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3.13 Union bounds and continuity

Set $B_1 = A_1$ and $B_i = A_i \setminus \bigcup_{j<i} A_j$. These events are disjoint, retain the same union, and satisfy $B_i \subseteq A_i$. Therefore

$$P\left(\bigcup_i A_i\right) = \sum_i P(B_i) \le \sum_i P(A_i).$$

The union bound holds for finite or countable families without independence.

For increasing $A_1 \subseteq A_2 \subseteq \cdots$, partition by successive differences. The first n terms sum to $P(A_n)$, giving

$$P(\bigcup_n A_n) = \lim_n P(A_n).$$

Apply this to complements of a decreasing sequence and use $P(A_n^c) = 1 - P(A_n)$ to get $P(\bigcap_n A_n) = \lim_n P(A_n)$.

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3.14 Equally likely outcomes and unequal value probabilities

Two fair independent coin tosses have ordered outcomes $\Omega = \{HH, HT, TH, TT\}$, each of probability $1/4$. For the head count X,

$$P(X = 0) = 1/4, \quad P(X = 1) = 1/2, \quad P(X = 2) = 1/4.$$

Value one has two preimages, so the three values are not each assigned $1/3$.

Likewise two independent fair dice have 36 equally likely ordered pairs. Sum two has one preimage, while sum seven has six. Sampling two distinct objects without replacement instead uses ordered distinct pairs, excluding diagonal (i, i).

Specify outcomes and masses first, then sum preimages of events or random-variable values.

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3.15 Conditioning restricts and renormalizes

For $P(B) > 0$, define

$$P(A \mid B) = P(A \cap B)/P(B).$$

Restrict outcomes to B and normalize by $P(B)$. This remains a probability model on B, preserving nonnegativity, unit total mass, and disjoint additivity.

The multiplication formula is

$$P(A \cap B) = P(B)P(A \mid B).$$

Repeated application gives the chain rule

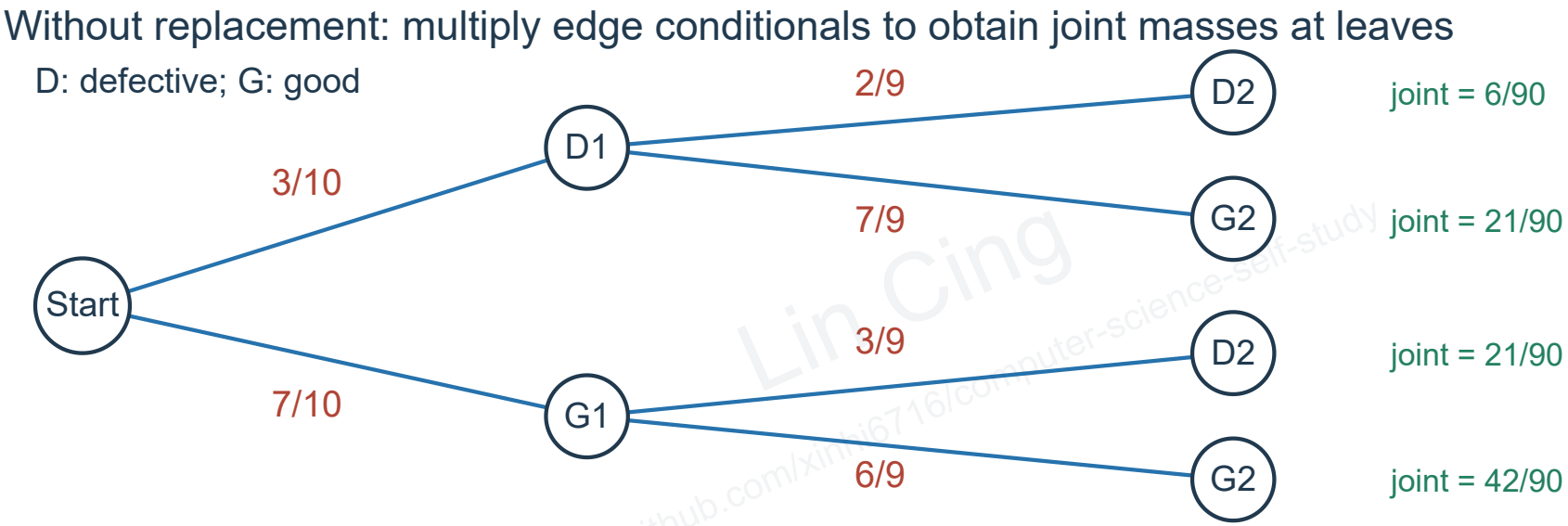
$$P(A_1 \cap \cdots \cap A_n) = P(A_1) \prod_{i=2}^{n} P(A_i \mid A_1 \cap \cdots \cap A_{i-1}).$$

Every conditioning prefix used must have positive probability. If one prefix has probability zero, the full intersection has probability zero; undefined conditionals need not be multiplied.

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3.16 A complete sampling-without-replacement tree

Among ten distinguishable components, three are defective and seven good. Draw two in order without replacement. Let D_i indicate a defect on draw i, and $G_i = D_i^c$.



For example, $P(D_1 \cap D_2) = \frac{3}{10}\frac{2}{9} = \frac{6}{90}$, rather than $(3/10)^2$. The two disjoint sources of a second-draw defect give

3.16 A complete sampling-without-replacement tree (continued)

$$P(D_2) = \frac{3}{10}\frac{2}{9} + \frac{7}{10}\frac{3}{9} = \frac{27}{90} = \frac{3}{10}.$$

Equal marginals $3/10$ do not imply independence, since $P(D_2 \mid D_1) = 2/9 \neq 3/10$.

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3.17 Partition proofs of total probability and Bayes

Let $B_1, B_2, \ldots$ partition Ω, with $P(B_i) > 0$ for every block used in conditioning. Since $A = \bigsqcup_i (A \cap B_i)$,

$$P(A) = \sum_i P(A \mid B_i) P(B_i).$$

Omit zero-probability blocks, whose intersections also have probability zero. Do not write an undefined conditional times zero.

For $P(A) > 0$ and fixed j,

$$P(B_j \mid A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A \mid B_j) P(B_j)}{\sum_i P(A \mid B_i) P(B_i)}.$$

Bayes' denominator includes all disjoint sources of A. Keeping only the numerator's class incorrectly forces the posterior to one.

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3.18 A Bayes example and its denominator

Assume the model $P(D) = 0.02, P(+ \mid D) = 0.95, P(+ \mid D^c) = 0.10$. Then

$$P(+) = 0.95(0.02) + 0.10(0.98) = 0.117,$$

$$P(D \mid +) = 0.019/0.117 = 19/117 \approx 16.24\%.$$

Multiplying these probabilities by 10,000 gives expected frequencies, not additional observations.

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3.18 A Bayes example and its denominator (continued)

Bayes: expected counts for 10,000 people under the assumed test model

Test result	D: disease	D^c: no disease	Total
Positive +	190	980	1170
Negative −	10	8820	8830
Total	200	9800	10000

Given positive: green row

P(D|+) = 190/1170

= 19/117 ≈ 16.24%

P(+|D) = 190/200 = 95%

Both numerators are 190. P(D|+) divides by 1170 positives; P(+|D) divides by 200 disease cases.

The low prior makes the non-D population large. Expected false positives number 980, exceeding the 190 true positives. The 95% is $P(+ \mid D)$, rather than $P(D \mid +)$.

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3.19 Independence differs from disjointness

Events are independent iff $P(A \cap B) = P(A)P(B)$. For $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$. The product definition also applies to zero-probability events.

Disjoint events with positive probabilities cannot be independent: their intersection has probability zero but the product is positive. Independent events may overlap substantially. Independence also preserves complements:

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A)(1 - P(B)).$$

Swap roles and complement again to obtain all four complement combinations.

In the uniform 30-integer model, $P(A \cap B) = 5/30 = (15/30)(10/30)$, so multiples of two and three are independent despite nonempty intersection.

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3.20 Pairwise independence need not be mutual

Mutual independence of $A_1, \ldots, A_n$ requires, for every nonempty index subset J,

$$P(\bigcap_{j \in J} A_j) = \prod_{j \in J} P(A_j).$$

Pairwise independence checks only size-two subsets.

On uniform $\Omega = \{00, 01, 10, 11\}$, let A mean first bit one, B second bit one, and C unequal bits.

Outcome	00	01	10	11
A	0	0	1	1
B	0	1	0	1
C	0	1	1	0

Each has probability $1/2$, and each pair intersects in one outcome, probability $1/4$. They are pairwise independent. The triple intersection is empty, with $0 \neq 1/8$, so mutual independence fails.

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3.21 Random variables and induced distributions

A discrete random variable is a function $X : \Omega \to \mathbb{R}$. The event $\{X = x\}$ is its preimage $\{\omega : X(\omega) = x\}$. Nonempty fibers partition Ω, giving

$$p_X(x) = P(X = x) \geq 0, \qquad \sum_x p_X(x) = 1.$$

Different functions may have the same distribution yet different joint behavior with other variables.

For a fair bit B, set $X = B, Y = 1 - B$. Each is uniform on $\{0, 1\}$, but they are always opposite and dependent. An additional independent fair bit Z has a different joint distribution with X.

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3.22 Expectation and LOTUS by grouping

If $E|X| = \sum_x |x| p_X(x) < \infty$, then

$$E[X] = \sum_x x p_X(x) = \sum_{\omega \in \Omega} X(\omega) p_\omega.$$

Group equal-value fibers to pass between forms. Absolute convergence permits rearrangement. A nonnegative variable may have expectation $+\infty$, but formulas involving subtraction here assume the needed expectations finite.

For $E|g(X)| < \infty$, the same grouping yields

$$E[g(X)] = \sum_x g(x) p_X(x).$$

This is LOTUS, the law of the unconscious statistician. No separate distribution for $g(X)$ is needed.

An expectation need not be a possible or most likely value. A fair $X \in \{0, 1\}$ has mean $1/2$, while $E[X^2] = 1/2 \neq (E[X])^2 = 1/4$.

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3.23 Linearity needs no independence

For $E|X|, E|Y| < \infty$, sum on the same space:

$$\begin{aligned} E[aX + bY] &= \sum_{\omega} (aX(\omega) + bY(\omega)) p_{\omega} \\ &= aE[X] + bE[Y]. \end{aligned}$$

No factorization of joint masses occurs, so independence is unnecessary. Induction extends this to finite sums.

For event indicator I_A, $E[I_A] = P(A)$. Thus a count $N = \sum_i I_{A_i}$ has $E[N] = \sum_i P(A_i)$, even with dependent events.

The equality $E[XY] = E[X]E[Y]$ needs extra conditions. Discrete independence permits factoring $p_{X,Y}(x, y) = p_X(x) p_Y(y)$ and separating the product sum.

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3.24 Variance and covariance

For $E[X^2] < \infty$ and $\mu = EX$,

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2.$$

Squares give nonnegativity. In a finite discrete distribution, variance is zero exactly when every positive-mass value agrees. Also $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$.

For finite second moments,

$$\operatorname{Cov}(X, Y) = E[(X - EX)(Y - EY)] = E[XY] - EX\,EY.$$

Expanding a square gives

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

Independence gives zero covariance; the converse generally fails. Linearity of expectation does not justify unconditional variance addition.

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3.25 Two variance calculations for one PMF

Let masses at 0,1,2 be $c, 2c, 3c$. Normalization gives $6c = 1$, so

$$EX = 0(1/6) + 1(2/6) + 2(3/6) = 4/3,$$
$$E[X^2] = 0 + 1(2/6) + 4(3/6) = 7/3.$$

Thus variance is $7/3 - 16/9 = 5/9$. Direct centering gives

$$\frac{1}{6}\left(-\frac{4}{3}\right)^2 + \frac{2}{6}\left(-\frac{1}{3}\right)^2 + \frac{3}{6}\left(\frac{2}{3}\right)^2 = \frac{16 + 2 + 12}{54} = \frac{5}{9}.$$

The mean $4/3$ lies outside the support, which does not prevent computing squared deviations from it.

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3.26 Hypergeometric sampling without replacement

A population of N distinguishable objects contains K defects. Uniformly sample an n-subset without replacement, and let X count defects. Then

$$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}},\quad \max(0,n-(N-K))\le k\le \min(n,K).$$

Choose defective and good identities separately in the numerator. Vandermonde proves normalization. Assume $N\ge 1, 0\le K\le N, 0\le n\le N$.

For $N=10, K=3, n=2$:

k	0	1	2
$P(X=k)$	$21/45=7/15$	$21/45=7/15$	$3/45=1/15$

Hence $EX=3/5, E[X^2]=11/15, \mathrm{Var}(X)=28/75$. Dependence makes this different from Binomial $(2,3/10)$.

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3.27 Negative covariance without replacement

For the same ten objects with three defects, define position indicators I_1, I_2. Each has expectation $3/10$, so $EX = 3/5$ by linearity, without independence. However,

$$E[I_1 I_2] = \frac{3}{10}\frac{2}{9} = \frac{1}{15}, \qquad \operatorname{Cov}(I_1, I_2) = \frac{1}{15} - \frac{9}{100} = -\frac{7}{300}.$$

Each indicator has variance $21/100$, so

$$\operatorname{Var}(X) = 2\frac{21}{100} + 2\left(-\frac{7}{300}\right) = \frac{28}{75}.$$

Drawing one defect reduces the remaining defect proportion. This creates negative covariance and reduces count variance without changing the identical marginal defect rates at the two positions.

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3.28 Bernoulli and binomial assumptions

A Bernoulli(p) variable $I \in \{0, 1\}$ has $P(I = 1) = p$, mean p, and variance $p(1 - p)$.

Suppose $I_1, \ldots, I_n$ are mutually independent, so every binary outcome tuple's joint mass factors, and all have the same $p \in [0, 1]$. Set $X = \sum I_i$. Each size-k set of success positions has probability $p^k(1 - p)^{n-k}$, with $\binom{n}{k}$ such sets. Hence

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

This is Binomial(n, p), $n \geq 0$. Interpret $p = 0, 1$ by the deterministic endpoint distributions.

Linearity and independence eliminating cross covariances give

$$EX = np, \qquad \operatorname{Var}(X) = np(1 - p).$$

Equal marginals alone, or even pairwise independence alone, do not guarantee this full PMF.

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3.29 Geometric probabilities and tails

Repeat independent Bernoulli(p) trials with $0 < p \leq 1$. The first-success trial T has support starting at one:

$$P(T = t) = (1-p)^{t-1}p, \quad t = 1, 2, \ldots.$$

The first $t-1$ trials fail and the next succeeds. Writing $q = 1 - p < 1$, total mass is $p \sum_{j \geq 0} q^j = 1$.

The tail is $P(T > k) = q^k$, requiring k initial failures. The pointwise identity $T = \sum_{k \geq 0} \mathbf{1}_{\{T > k\}}$ gives

$$ET = \sum_{k \geq 0} P(T > k) = \sum_{k \geq 0} q^k = 1/p.$$

Nonnegative terms permit exchanging sums. At $p = 0$, success never occurs; this is not a normalized geometric distribution on finite positive integers.

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3.30 Geometric second moments

Pointwise, $T^2 = \sum_{k=0}^{T-1} (2k+1)$, giving

$$E[T^2] = \sum_{k \geq 0} (2k+1) q^k.$$

For $S = \sum_{k \geq 0} k q^k$, subtracting qS gives $S - qS = \sum_{k \geq 1} q^k = q/(1-q)$, so $S = q/(1-q)^2$. Therefore

$$E[T^2] = \frac{2q}{(1-q)^2} + \frac{1}{1-q} = \frac{2-p}{p^2},$$

$$\operatorname{Var}(T) = \frac{1-p}{p^2}.$$

For $p = 1/4$, mean is four and variance twelve. For $p = 1$, $T = 1$, with mean one and variance zero. Counting failures before success instead gives $T - 1$, whose mean is one less.

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3.31 Indicator proofs of Markov and Chebyshev

For nonnegative Y and $a > 0$, $Y \geq a\mathbf{1}_{\{Y \geq a\}}$ pointwise, hence

$$P(Y \geq a) \leq E[Y]/a.$$

This is Markov's inequality. Finite expectation gives a useful finite bound, which can also be capped at one.

For X with finite variance, take $Y = (X - EX)^2, a = t^2, t > 0$:

$$P(|X - EX| \geq t) \leq \mathrm{Var}(X)/t^2.$$

Chebyshev needs no special distribution or independence. It uses a nonnegative square and finite second moment. A right side above one remains valid but is uninformative.

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3.32 Joint, marginal, and conditional PMFs

For discrete X, Y, let $p(x, y) = P(X = x, Y = y)$. Summing disjoint fibers gives

$$p_X(x) = \sum_y p(x, y), \qquad p_Y(y) = \sum_x p(x, y).$$

For $p_Y(y) > 0$, define $p_{X|Y}(x|y) = p(x, y)/p_Y(y)$ and $E[X|Y = y] = \sum_x x\, p_{X|Y}(x|y)$, assuming the required absolute sums finite.

Grouping by the partition induced by Y gives total expectation:

$$E[X] = \sum_y E[X|Y = y] p_Y(y).$$

No independence is needed. Zero-probability values need no conditional definition and are omitted from the sum.

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3.33 Why the total-variance cross term vanishes

Assume $E[X^2] < \infty$, and write $m(y) = E[X|Y = y]$. Conditional on $Y = y$,

$$X - EX = (X - m(y)) + (m(y) - EX).$$

After squaring and taking conditional expectation, the cross term is

$$2(m(y) - EX)E[X - m(y)|Y = y] = 0.$$

Thus the conditional squared deviation is $\mathrm{Var}(X|Y = y) + (m(y) - EX)^2$. Average over Y:

$$\mathrm{Var}(X) = E[\mathrm{Var}(X|Y)] + \mathrm{Var}(E[X|Y]).$$

The first term averages within-group variances. The second measures variation of group means. Averaging only within-group variances misses the second contribution.

All terms are finite: nonnegative conditional variance gives $m(y)^2 \le E[X^2|Y = y]$. Averaging yields $E[m(Y)^2] \le E[X^2] < \infty$, avoiding undefined infinity subtraction.

3.34 Connecting proofs and calculations

Prove set identities through arbitrary-element predicates; check Venn counts through nonempty-region partitions. Probability then assigns outcome masses. Geometry or the number of possible values cannot establish equal likelihood.

Check positive denominators and partitions before conditioning. Specify event independence, pairwise independence, or mutual independence. Expectation is linear without independence; variance retains covariance. Every distribution requires its stated support and experiment.

Separate exercise and solution PDFs give new cases with complete derivations. Finite enumeration checks specified models and does not replace general axiomatic proofs.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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